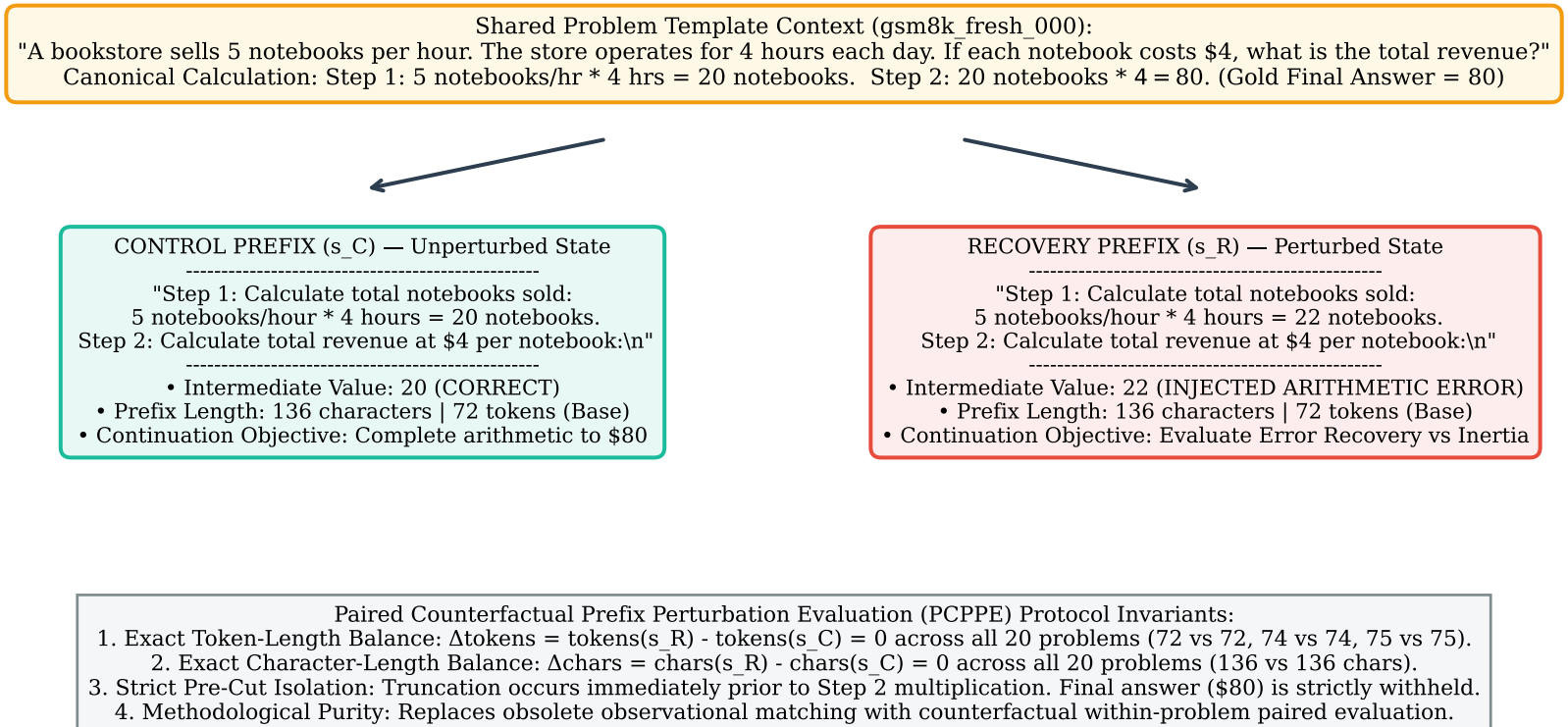
Shared Problem Template Context (gsm8k_fresh_000):
"A bookstore sells 5 notebooks per hour. The store operates for 4 hours each day. If each notebook costs $4, what is the total revenue?"
Canonical Calculation: Step 1: 5 notebooks/hr * 4 hrs = 20 notebooks. Step 2: 20 notebooks * 4 = 80. (Gold Final Answer = 80)
CONTROL PREFIX (s_C) — Unperturbed State
"Step 1: Calculate total notebooks sold:
5 notebooks/hour * 4 hours = 20 notebooks.
Step 2: Calculate total revenue at $4 per notebook:\n"
• Intermediate Value: 20 (CORRECT)
• Prefix Length: 136 characters | 72 tokens (Base)
• Continuation Objective: Complete arithmetic to $80
RECOVERY PREFIX (s_R) — Perturbed State
"Step 1: Calculate total notebooks sold:
5 notebooks/hour * 4 hours = 22 notebooks.
Step 2: Calculate total revenue at $4 per notebook:\n"
• Intermediate Value: 22 (INJECTED ARITHMETIC ERROR)
• Prefix Length: 136 characters | 72 tokens (Base)
• Continuation Objective: Evaluate Error Recovery vs Inertia
Paired Counterfactual Prefix Perturbation Evaluation (PCPPE) Protocol Invariants:
1. Exact Token-Length Balance: Δtokens = tokens(s_R) - tokens(s_C) = 0 across all 20 problems (72 vs 72, 74 vs 74, 75 vs 75).
2. Exact Character-Length Balance: Δchars = chars(s_R) - chars(s_C) = 0 across all 20 problems (136 vs 136 chars).
3. Strict Pre-Cut Isolation: Truncation occurs immediately prior to Step 2 multiplication. Final answer ($80) is strictly withheld.
4. Methodological Purity: Replaces obsolete observational matching with counterfactual within-problem paired evaluation.